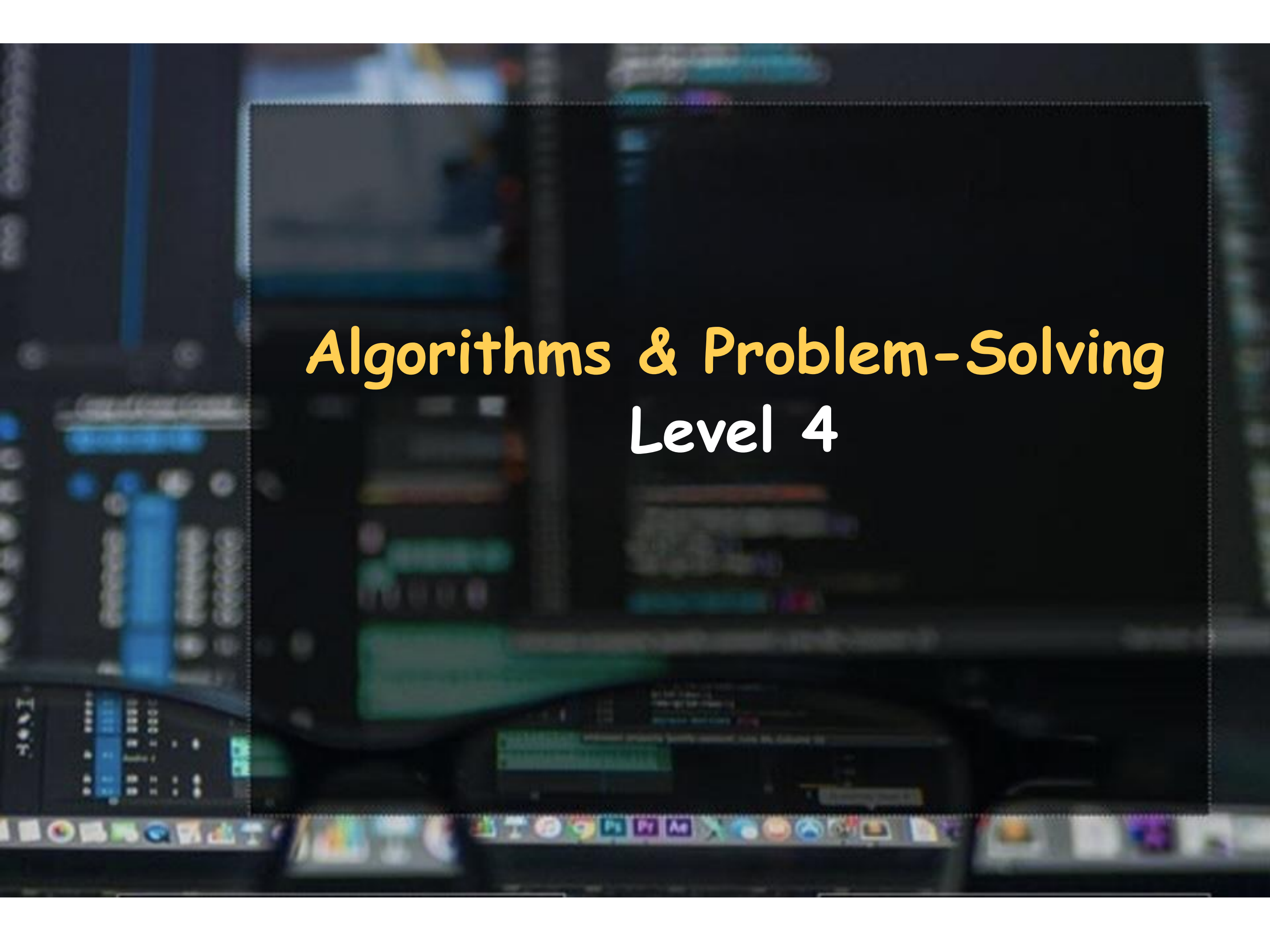
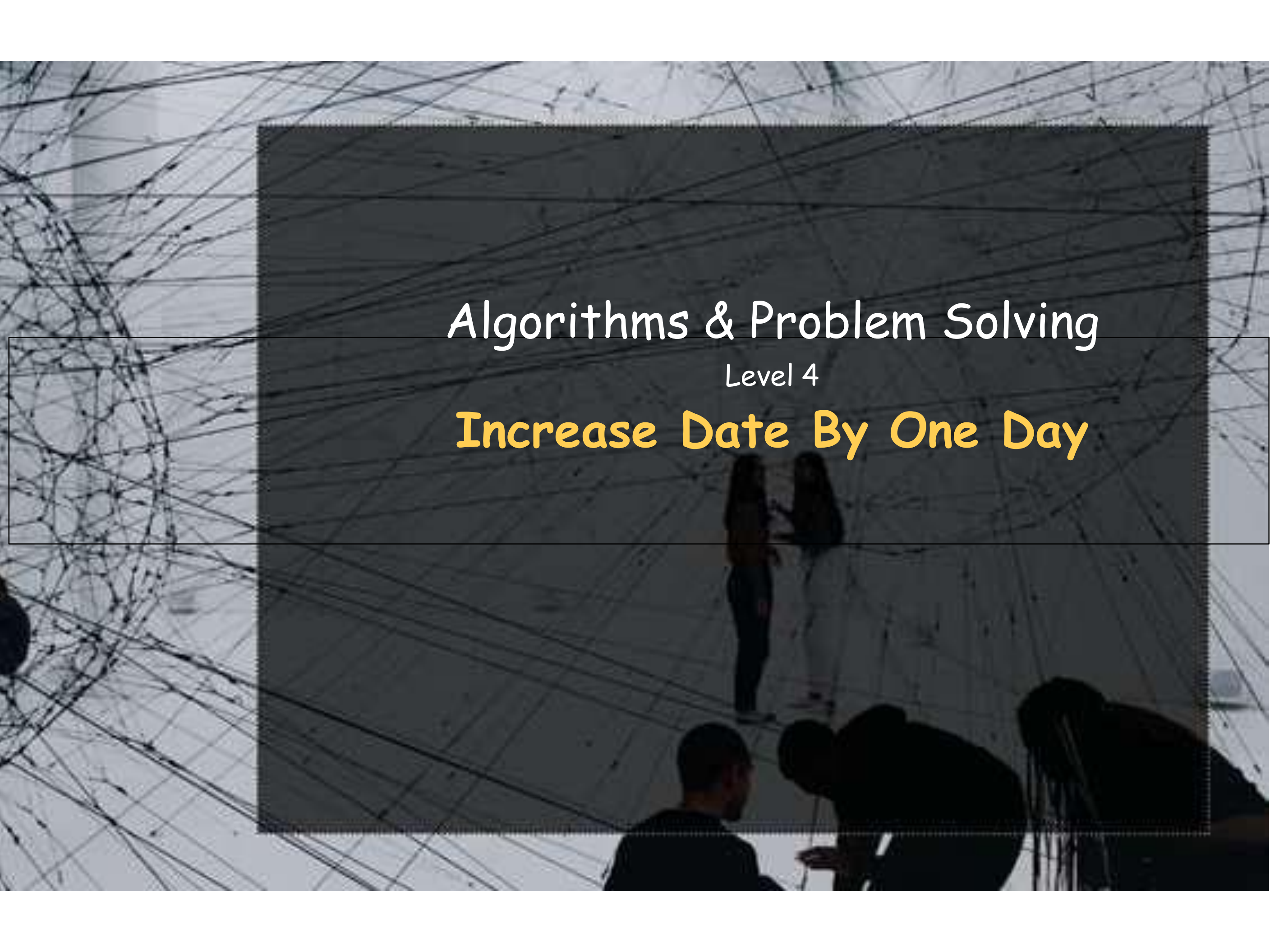




# Algorithms & Problem-Solving Level 4



# Algorithms & Problem Solving

Level 4

## Increase Date By One Day

# Definition

-  The goal is a program that reads the date (day, month, year) from the user, then adds this date by just one day and displays the result on the screen.



# Explanation

- The goal is to correctly handle the changing days, especially when transitioning from the end of one month to the beginning of the next, or from the end of one year to the next.
- The program relies on a set of organized functions to divide tasks such as reading the date, checking for leap years, and calculating the number of days in a month.

# Procedures

- ① Defining the sDate structure to store the following information:
  - Year — Year
  - Month — Month
  - Day — Day
- ② Reading the date from the user using the following functions:
  - ReadDay()i
  - ReadMonth()i
  - ReadYear()i
- ③ Checking for leap years:
  - IsLeapYear()i
  - Returns true if the year is a leap year, i.e., divisible by 4 but not by 100, or divisible by 400
- ④ Determining the number of days in each month:
  - NumberOfDaysInAMonth()i
  - Specifies the number of days according to the month and year (February = 28 or 29 in a leap year)
- ⑤ Checking the last day of the month:
  - IsLastDayInMonth()i
  - Checks if the current day is the last day of the month Day of the Month
- ⑥ Checking the Last Month of the Year:
  - IsLastMonthInYear()i
  - Returns true if the month is December (12)

7 Increasing the Date by One Day:

- IncreaseDateByOneDay()

- If it's the last day of the month and the last day of the year:

  - The year is incremented by +1, and the month and day are set to 1

- If it's only the last day of the month (not the year):

  - The month is incremented by +1, and the day is set back to 1

- Otherwise:

  - Only the day is incremented by +1

8 Displaying the Final Result: ■ Prints the new date after incrementing by one day



## Example:

Please enter a Day? 31

Please enter a Month? 12

Please enter a Year? 2022

Date after adding one day is: 1/1/2023



## Note:

- The program accurately handles transitions between months and years, including leap years.
- The code is extensible to accommodate more days by using a loop that calls the same function multiple times.

## Explanation:

- The program links the logic of the *Gregorian* calendar with its mathematical rules:
  - When the number of days in a month ends, the counting starts again and the month or year increases as needed.

## **Warning:**

- If a month outside the range (1-12) or an invalid day is entered, the program will not give an accurate result because it does not have input error handling.

## Summary

- ✓ The program accurately advances any date by one day.
- ✓ It takes into account the last days of months and leap years.

Thank you 😊